Self-Distillation with Frequency-Adaptive 2D Rotary Position Embeddings, Spatial-Semantic Sparse Attention, and Verifiable Group Relative Policy Post-Training for Vision Transformers

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November 1, 2026

Abstract

Self-supervised knowledge distillation without labels, notably pioneered by DINOv1, enables Vision Transformers (ViTs) to learn semantic representations exhibiting emergent scene segmentation and clustering capabilities. However, standard ViTs in self-supervised pipelines rely predominantly on learnable absolute positional embeddings (APEs) and dense multi-head self-attention. This classical formulation suffers from three fundamental limitations: (1) poor geometric generalization when processing multi-scale image crops due to rigid bicubic interpolation artifacts, (2) quadratic computational complexity paired with diffuse attention dispersion, where uninformative background clutter (e.g., sky, ground) pollutes the central `[CLS]` representation, and (3) the absence of an explicit spatial post-training mechanism to suppress residual background activations without human annotations. In this work, we propose a two-stage self-supervised framework for Vision Transformers. In **Stage 1 (Architectural Pretraining)**, we introduce: (i) **Frequency-Adaptive 2D Rotary Position Embedding (2D RoPE)**, which decomposes spatial coordinate tokens along frequency spectra to balance local high-frequency boundary details with scale-invariant low-frequency global topology, and (ii) **Spatial-Semantic Sparse Attention**, which unifies local spatial window connectivity ($K_{\text{local}} \times K_{\text{local}}$), dynamic global semantic Top-K routing, and bidirectional `[CLS]` token dispatch while strictly preserving standard Softmax probability normalization. In **Stage 2 (Self-Supervised RLVR Post-Training)**, we introduce **Unbiased Group Relative Policy Optimization (Dr. GRPO)** over a shared-prefix policy wrapper that unlocks only the final two ViT blocks and a lightweight residual patch-gating head (8.27% of wrapper parameters). Without using any class labels or neural reward models, Stage 2 optimizes verifiable visual rewards combining horizontal-flip Geometric Equivariance IoU, spectral TokenCut foreground compactness, and a target sparsity prior ($\rho_{\text{target}} = 0.35$) regularized by exact analytical Bernoulli KL divergence against the frozen Stage-1 reference model. Pretrained without labels on **ImageNet-100** for **190 epochs** on an **NVIDIA H100 GPU** and evaluated on **CIFAR-10**, our Stage-1 architecture achieves **75.97%** k-NN ($k = 10$) Top-1 accuracy (+1.56% over 74.41%), **75.31%** k-NN ($k = 20$) Top-1 accuracy (+0.97% over 74.34%), and **83.98%** standardized Linear Probing Top-1 accuracy (+2.52% over 81.46%) with **zero parameter overhead** (21.67M backbone parameters, 9.20 GFLOPs). Furthermore, a rigorous 2×2 ablation study across the full CIFAR-10 benchmark demonstrates that Stage-2 Dr. GRPO post-training dramatically boosts unsupervised Foreground Mask IoU by **+20.10%** on Vanilla DINOv1 ($58.88\% \rightarrow 78.98\%$, CorLoc $79.93\% \rightarrow 99.91\%$) and **+16.99%** on RoPE+Sparse DINOv1 ($55.45\% \rightarrow 72.44\%$, CorLoc $69.26\% \rightarrow 98.17\%$), while further improving downstream k-NN and Linear Probe accuracy.

Keywords: Self-Supervised Learning, Knowledge Distillation, DINOv1, Vision Transformer, Rotary Position Embedding (RoPE), Sparse Attention, Group Relative Policy Optimization (Dr. GRPO), Reinforcement Learning with Verifiable Rewards (RLVR).

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1 Introduction

Self-Supervised Learning (SSL) has reshaped modern representation learning across computer vision and natural language processing. In visual domains, self-distillation methods—particularly DINO (Distillation with no labels) [1]—have demonstrated that Vision Transformers (ViTs) [2] trained without human annotations naturally acquire explicit scene semantics, object boundaries, and feature linear separability superior to supervised baselines.

Despite these empirical successes, canonical ViT architectures employed in DINOv1 retain architectural vestiges from standard sequence models that are suboptimal for visual data:

1. **Geometric Brittleness of Absolute Positional Embeddings:** Standard ViTs add learnable 1D or 2D Absolute Positional Embeddings (APEs) to image patch tokens prior to Transformer blocks. In self-supervised multi-crop training [1,3], networks alternate between global crops (224×224, 14×14 patches) and local crops (96×96, 6×6 patches). Resizing static positional vectors via bicubic interpolation degrades relative spatial orientation, distorts high-frequency phase information, and ignores translation equivariance.

2. **Attention Dilution in Dense Softmax Attention:** Dense Multi-Head Self-Attention scales quadratically $\mathcal{O}(N^2)$ with the patch sequence length N. More critically, fully dense attention allows semantic tokens to disperse attention mass indiscriminately over extensive background regions (such as cloudy skies or uniform pastures). This leads to representational pollution of the `[CLS]` token, hindering downstream classification and fine-grained boundary segmentation.

3. **Residual Background Leakage After Global Self-Distillation:** Because the DINO cross-entropy objective operates solely on global `[CLS]` probability projections, pretraining lacks an explicit spatial verification signal at the patch level. Consequently, converged checkpoints still exhibit non-trivial attention mass on co-occurring background contexts, and standard fine-tuning requires labeled segmentation masks.

4. **High Compute Overhead in Self-Supervised Training:** State-of-the-art SSL pipelines frequently require massive GPU clusters (e.g., 8 to 64 V100/A100 GPUs), creating substantial barriers for individual researchers and academic laboratories constrained by consumer hardware (e.g., 8GB–16GB VRAM).

To address these challenges, we introduce a two-stage self-supervised Vision Transformer framework integrating geometric-sparse pretraining with label-free reinforcement policy post-training. Our primary contributions are summarized as follows:

- **Frequency-Adaptive 2D Rotary Position Embedding (2D RoPE):** We extend Rotary Position Embeddings [4] to continuous two-dimensional visual grids with frequency-aware coordinate adaptation. High-frequency dimensions capture fine-grained pixel boundaries using raw coordinates, while low-frequency bands generalize to varying resolutions via normalized coordinates.

- **Spatial-Semantic Sparse Softmax Attention:** We design a composite sparse attention mechanism combining a local 2D Chebyshev spatial window, a dynamic global semantic Top-K candidate selection, and bidirectional `[CLS]` routing. Additionally, we provide a multi-scale 2D sliding-window variant (`MultiScale2DWindowAttention`) grouping heads across 3×3, 5×5, 7×7, and full global receptive fields. Both designs retain exact Softmax properties without probability truncation.

- **Stage-2 Self-Supervised Post-Training via Dr. GRPO:** We formulate foreground patch selection in pretrained DINO backbones as a Reinforcement Learning with Verifiable Rewards (RLVR) problem using Unbiased Group Relative Policy Optimization (`DrGRPOLoss`) [18, 19]. By sampling groups of binary patch masks from a shared-prefix policy wrapper (`DINOGRPOPolicyWrapper`) and optimizing a purely self-supervised reward—combining horizontal-flip Geometric Equivariance IoU, spectral TokenCut compactness [21], and target sparsity—we sharpen object localization and feature quality without human labels.

- **Modular Registry and Declarative Configuration Builder:** We implement a decoupled, clean PyTorch architecture featuring a dynamic component registry (`MODULES`, `BACKBONES`, `HEADS`, `MODELS`, `LOSSES`, `OPTIMIZERS`, `TRANSFORMS`) allowing flexible network and training pipeline instantiation via declarative YAML configurations.
- **Comprehensive Stage-1 and 2×2 Stage-2 Empirical Validation:** Pretrained without labels on **ImageNet-100** for **190 epochs** on an **NVIDIA H100 GPU** and evaluated on **CIFAR-10**, our Stage-1 ViT-Small/16 variant achieves **75.97%** k-NN ($k = 10$) Top-1 accuracy (+1.56%), **75.31%** k-NN ($k = 20$) Top-1 accuracy (+0.97%), and **83.98%** Linear Probing Top-1 accuracy (+2.52% over 81.46%) with zero parameter overhead. Furthermore, our 2×2 ablation study (Architecture $\times$ Stage-2 Dr. GRPO) on full CIFAR-10 demonstrates gains of **+16.99% to +20.10%** in unsupervised Foreground Mask IoU (reaching up to **99.91%** CorLoc) alongside consistent downstream classification improvements.

2 Related Work

2.1 Self-Supervised Visual Representation Learning

Self-supervised visual learning paradigms have evolved from contrastive methods like MoCo [5, 6] and SimCLR to non-contrastive knowledge distillation frameworks like BYOL [7], SwAV [3], and DINO [1]. DINO interprets self-supervised learning as knowledge distillation with no labels: a student network is trained to predict the output distribution of a dynamically momentum-updated teacher network. By combining multi-crop data augmentation with centering and sharpening operations on the teacher's probability distribution, DINO avoids trivial representation collapse without negative sample pairs. DINOv2 [8] scaled this approach to billion-parameter datasets with register tokens [9]. In contrast, our focus is optimizing architectural inductive biases and positional geometry under constrained compute budgets.

2.2 Positional Encodings in Vision Transformers

The canonical Vision Transformer [2] maps non-overlapping image patches to vectors and adds 1D learnable absolute positional embeddings. While functional for fixed-size classification, APE lacks relative translation invariance. Relative Positional Encodings (RPE) introduce distance biases directly into the attention logits. Rotary Position Embedding (RoPE) [4] was developed for autoregressive language modeling, encoding relative token distances through complex vector rotations. Recent vision adaptations extend RoPE to 2D image coordinates; however, existing methods apply uniform scaling across all channels, overlooking the disparate multi-scale demands of high-frequency local textures versus low-frequency global layouts.

2.3 Sparse and Efficient Attention

Dense multi-head attention suffers from $\mathcal{O}(N^2)$ time and memory scaling. Sparse Transformers [10], Longformer [11], and BigBird [12] demonstrated that sparse connectivity graphs combining local sliding windows and global routing achieve competitive representational power in long text sequences. Swin Transformer [13] pioneered shifted local windows for vision backbones. In our framework, we specifically tailor sparse connectivity for self-distillation: we blend local spatial proximity with semantic Top-K content routing and full `[CLS]` token routing, preserving exact Softmax normalization and zero parameter overhead.

2.4 Reinforcement Learning with Verifiable Rewards and GRPO

Group Relative Policy Optimization (GRPO) [18] eliminated the costly value critic network of Proximal Policy Optimization (PPO) by estimating baselines directly from groups of sampled action rollouts.

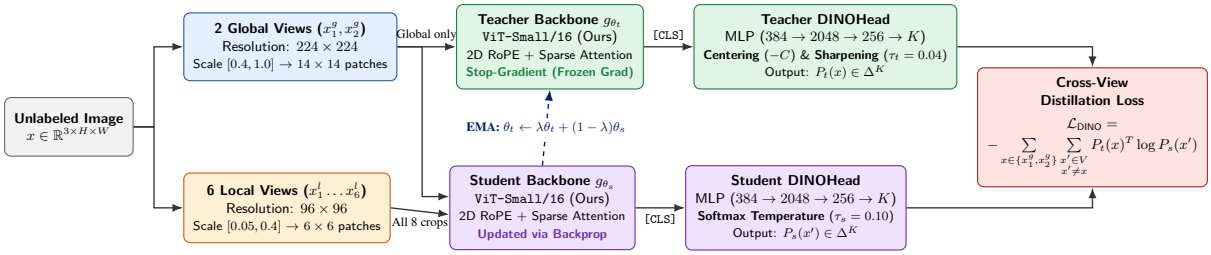


Figure 1: Overview of the DINO self-distillation framework with Multi-Crop augmentation. From a single unlabeled image, 2 global views (224×224) and 6 local views (96×96) are generated. The Teacher processes only the global views, whereas the Student processes all 8 views. Both networks utilize our enhanced `ViT-Small/16` backbone while teacher weights are updated via Exponential Moving Average (EMA).

Recently, Dr. GRPO [19] identified that dividing group-centered rewards by the intra-group standard deviation introduces variance-dependent gradient bias and length distortion, advocating instead for unbiased mean-centered advantages paired with length-normalized log-probabilities. Concurrently, DAPO [20] introduced asymmetric clipping (*Clip-Higher*, $\epsilon_{\text{high}} > \epsilon_{\text{low}}$) to prevent premature entropy collapse on high-advantage exploratory actions. While RLVR has predominantly targeted autoregressive language reasoning, we adapt Dr. GRPO to self-supervised visual representation learning by treating last-layer `[CLS]` attention and a lightweight residual patch gate as a multi-variate Bernoulli policy over 2D image patches, optimized via label-free geometric equivariance and graph-spectral TokenCut [21] verifiable rewards.

3 Methodology

3.1 DINO Self-Distillation Framework

The DINO framework (illustrated in Figure 1) consists of two neural networks sharing identical structural backbones: a student g_{θ_s} parameterized by weights θ_s and a teacher g_{θ_t} parameterized by θ_t. Given an input image x, multi-crop augmentation generates a set of views $V = \{x_1^g, x_2^g, x_1^l, \dots, x_M^l\}$, containing two global crops x_1^g, x_2^g of resolution 224×224 (covering scale $[0.4, 1.0]$) and $M = 6$ local crops x_m^l of resolution 96×96 (covering scale $[0.05, 0.4]$).

The teacher only processes the two global views, while the student processes all global and local views. The networks map views to K-dimensional probability distributions via projection heads h:

$$P_s(x)^{(k)} = \frac{\exp(g_{\theta_s}(x)^{(k)}/\tau_s)}{\sum_{j=1}^{K}\exp(g_{\theta_s}(x)^{(j)}/\tau_s)} \tag{1}$$

$$P_t(x)^{(k)} = \frac{\exp((g_{\theta_t}(x)^{(k)} - C^{(k)})/\tau_t)}{\sum_{j=1}^{K}\exp((g_{\theta_t}(x)^{(j)} - C^{(j)})/\tau_t)} \tag{2}$$

where τ_s and τ_t are the student and teacher temperature hyperparameters, and $C \in \mathbb{R}^K$ is a dynamic centering vector. The student parameters θ_s are optimized by minimizing the cross-entropy loss over all cross-view pairs:

$$\mathcal{L}_{\text{DINO}} = \sum_{x\in\{x_1^g,x_2^g\}}\sum_{\substack{x'\in V\\ x'\neq x}} -P_t(x)^T \log P_s(x') \tag{3}$$

To avoid collapse without negative pairs, DINO applies two counterbalancing mechanisms:

1. **Sharpening:** Setting a small teacher temperature $\tau_t < \tau_s$ prevents the output distribution from degenerating into a uniform collapse.

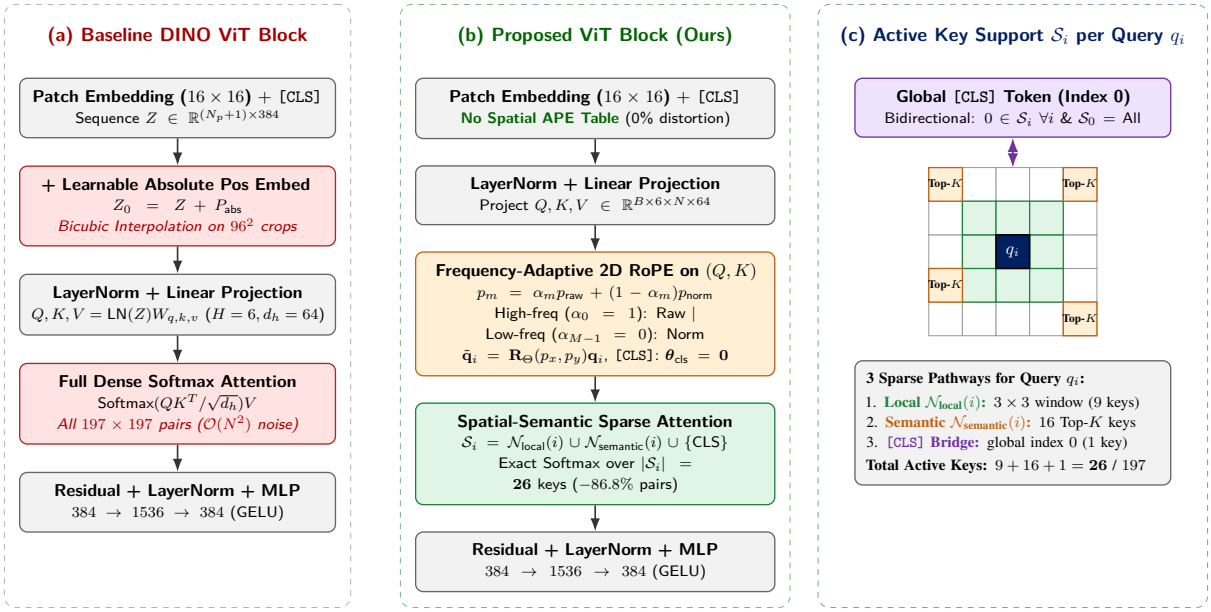


Figure 2: Detailed architectural comparison between **(a)** the standard DINO Vision Transformer block (relying on static Absolute Positional Embeddings resized via bicubic interpolation and 197×197 dense attention) and **(b)** our proposed ViT block integrating Frequency-Adaptive 2D RoPE and Spatial-Semantic Sparse Softmax Attention. **(c)** Illustration of the sparse attention support $\mathcal{S}_i$ for a spatial query q_i, combining a 3×3 local window, 16 dynamic semantic Top-K tokens, and bidirectional [CLS] routing.

2. **Centering:** Subtracting an Exponential Moving Average (EMA) vector C prevents one dimension from dominating:

$$C \leftarrow m \cdot C + (1 - m)\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(x_i) \tag{4}$$

with momentum coefficient $m \in [0, 1]$.

The teacher network is updated via an Exponential Moving Average of student weights:

$$\theta_t \leftarrow \lambda\theta_t + (1 - \lambda)\theta_s \tag{5}$$

with λ scheduled via a cosine curve from 0.996 to 1.0 across pretraining.

3.2 Frequency-Adaptive 2D Rotary Position Embedding

In standard 1D RoPE [4], an embedding vector $\mathbf{x} \in \mathbb{R}^d$ at sequence index p is rotated by an orthogonal matrix $\mathbf{R}_{\Theta,p}$. For a 2D patch grid $H_g \times W_g$, each patch token is indexed by spatial coordinates (p_y, p_x) where $p_y \in [0, H_g - 1]$ and $p_x \in [0, W_g - 1]$.

Head Dimension Partitioning: For an attention head of dimension d_h, we allocate half the channels to the horizontal axis ($d_x = d_h/2$) and half to the vertical axis ($d_y = d_h/2$). The number of distinct frequency bands per spatial dimension is $M = d_x/2 = d_h/4$. The base frequencies ω_m are given by:

$$\omega_m = b^{-2m/d_x}, \quad m \in \{0, 1, \ldots, M - 1\} \tag{6}$$

with base $b = 10000.0$.

Frequency-Adaptive Coordinate Blending: Standard RoPE implementations either utilize raw integer coordinates p_{raw} or normalized coordinates $p_{\text{norm}} = p_{\text{raw}} \cdot (S_{\text{ref}}/S_{\text{grid}})$. We observe that high-frequency dimensions encode local, fine-grained structural textures that depend strictly on pixel distances, whereas low-frequency dimensions represent global topological arrangement, which should remain scale-invariant across multi-crop transformations. We formalize this through a frequency-dependent interpolation coefficient α_m:

$$p_m = \alpha_m \cdot p_{\text{raw}} + (1 - \alpha_m) \cdot p_{\text{norm}} \tag{7}$$

where α_m is monotonically decreased across frequency bands:

$$\alpha_m = 1.0 - \frac{m}{M-1}, \quad m \in \{0, \ldots, M-1\} \tag{8}$$

For $m = 0$ (highest frequency), $\alpha_0 = 1.0$ (pure raw coordinates). For $m = M - 1$ (lowest frequency), $\alpha_{M-1} = 0.0$ (pure normalized coordinates).

Rotary Application to Queries and Keys: The 2D rotation angles along each axis are computed as $\theta_{x,m} = p_{x,m} \cdot \omega_m$ and $\theta_{y,m} = p_{y,m} \cdot \omega_m$. Each angle is paired into 2×2 block diagonal rotation matrices:

$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{9}$$

The composite transformation matrix $\mathbf{R}_\Theta(p_x, p_y)$ rotates the projected Query $\mathbf{q}_i$ and Key $\mathbf{k}_j$ vectors:

$$\tilde{\mathbf{q}}_i = \mathbf{R}_\Theta(p_{x,i}, p_{y,i})\mathbf{q}_i, \quad \tilde{\mathbf{k}}_j = \mathbf{R}_\Theta(p_{x,j}, p_{y,j})\mathbf{k}_j \tag{10}$$

Crucially, the inner product inherits relative translation equivariance:

$$\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle = \mathbf{q}_i^T \mathbf{R}_\Theta(p_{x,i} - p_{x,j}, p_{y,i} - p_{y,j})\mathbf{k}_j \tag{11}$$

Treatment of the `[CLS]` Token: The special classification token `[CLS]` lacks a predefined spatial coordinate. To preserve its global semantic invariance, we assign it rotation angles $\boldsymbol{\theta}_{\text{cls}} = \mathbf{0}$, yielding $\mathbf{R}_\Theta(\text{cls}) = \mathbf{I}$, while retaining its learnable 1D class positional bias. Thus, query-key interactions involving `[CLS]` remain unrotated, preventing artificial spatial bias.

3.3 Spatial-Semantic Sparse Softmax Attention

Standard dense attention computes pairwise similarities across all N tokens, producing an $N \times N$ matrix. In our Spatial-Semantic Sparse Attention (`SpatialSemanticAttention`, Figure 2b–c), each query token i is constrained to attend only to an active subset of keys $\mathcal{S}_i \subset \{0, 1, \ldots, N-1\}$. We construct $\mathcal{S}_i$ by unifying three complementary pathways:

1. **Local Spatial Neighborhood ($\mathcal{N}_{\text{local}}(i)$):** For a patch i located at 2D grid coordinates (r_i, c_i), its local receptive field comprises all patches within Chebyshev window radius $R = \lfloor K_{\text{local}}/2 \rfloor$:

$$\mathcal{N}_{\text{local}}(i) = \{j \mid |r_i - r_j| \leq R \wedge |c_i - c_j| \leq R\} \tag{12}$$

   With $K_{\text{local}} = 3$, each patch directly connects to its immediate 8-neighborhood plus itself ($3 \times 3 = 9$ spatial tokens).

2. **Global Semantic Top-K ($\mathcal{N}_{\text{semantic}}(i)$):** To capture long-range semantic correlations outside the local window, we evaluate scaled dot-product similarities $\text{sim}(i, j) = (\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_j)/\sqrt{d_h}$ across non-local candidates $\mathcal{C}_i = \{j \mid j \notin \mathcal{N}_{\text{local}}(i), j \neq 0\}$, and select the K_{semantic} highest scoring tokens:

$$\mathcal{N}_{\text{semantic}}(i) = \text{argtop-}K_{\text{semantic}}\{\text{sim}(i, j) \mid j \in \mathcal{C}_i\} \tag{13}$$

   In our experiments, $K_{\text{semantic}} = 16$.

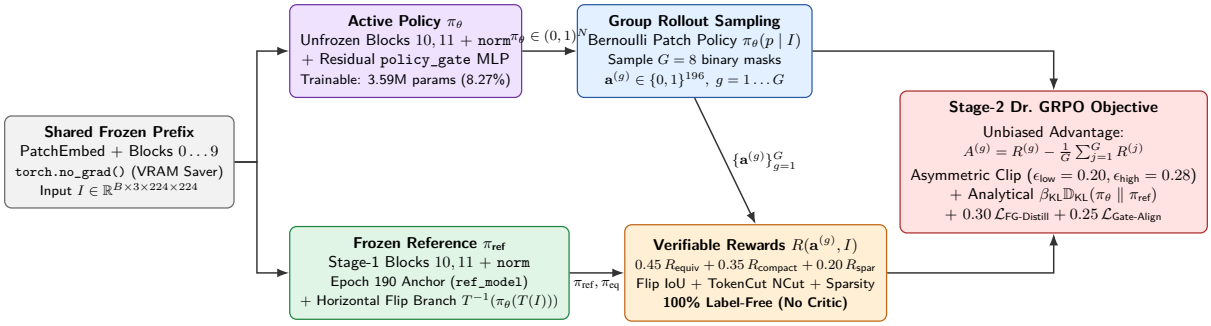


Figure 3: Overview of the Stage-2 Self-Supervised RLVR Post-Training pipeline (`DINOGRPOPolicyWrapper` and `DrGRPOLoss`). Early ViT blocks $(0 \dots 9)$ are frozen and shared in a single `no_grad` pass. The active policy unlocks only the last 2 blocks and a lightweight residual `policy_gate`, sampling $G = 8$ binary foreground masks per image and optimizing label-free verifiable visual rewards alongside exact Bernoulli KL regularization.

3. **Bidirectional `[CLS]` Routing:** The `[CLS]` token (index 0) serves as a global information bottleneck. We enforce full bidirectional connectivity:

$$\mathcal{S}_0 = \{0, 1, \dots, N-1\}, \quad 0 \in \mathcal{S}_i \quad \forall i \tag{14}$$

The complete active key index set for query token i is the union:

$$\mathcal{S}_i = \mathcal{N}_{\text{local}}(i) \cup \mathcal{N}_{\text{semantic}}(i) \cup \{0\} \tag{15}$$

Exact Softmax Probability Normalization: Rather than approximating attention through linear projections or sampling, we enforce exact Softmax normalization over the sparse support:

$$\mathbf{A}_{ij} = \begin{cases} \dfrac{\exp\left(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_j / \sqrt{d_h}\right)}{\sum_{l \in \mathcal{S}_i} \exp\left(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_l / \sqrt{d_h}\right)}, & j \in \mathcal{S}_i \\ 0, & j \notin \mathcal{S}_i \end{cases} \tag{16}$$

The attention output is obtained via sparse matrix multiplication:

$$\mathbf{O}_i = \sum_{j \in \mathcal{S}_i} \mathbf{A}_{ij} \mathbf{v}_j \tag{17}$$

This formulation guarantees that the attention weights for each query sum strictly to 1, avoiding numerical instabilities and gradient vanishing. Additionally, our framework supports a multi-scale sliding-window attention variant (`MultiScale2DWindowAttention`) that partitions the $H = 6$ heads into multi-scale local receptive fields (3×3, 5×5, 7×7 via fused 2D `F.unfold`) alongside global dense heads and a bidirectional `[CLS]` bridge.

3.4 Stage-2 Self-Supervised Post-Training via Dr. GRPO

While Stage-1 DINO self-distillation aligns global `[CLS]` distributions across views, it lacks an explicit patch-level objective to suppress residual background activations. To refine spatial object localization and representation quality without human annotations or neural reward models, we introduce a Stage-2 Reinforcement Learning with Verifiable Rewards (RLVR) post-training pipeline based on Unbiased Group Relative Policy Optimization (Dr. GRPO) [18, 19] (illustrated in Figure 3).

1. Shared-Prefix Policy Wrapper (`DINOGRPOPolicyWrapper`): Given a Stage-1 pretrained Vision Transformer backbone of depth $L = 12$ and $N = H_g \times W_g = 196$ spatial patches, we freeze the patch embedding and the first $L - B_{\text{unfrozen}} = 10$ Transformer blocks, unlocking only the final $B_{\text{unfrozen}} = 2$ blocks (`blocks[10..11]`), the final LayerNorm, and a lightweight residual gating MLP `policy_gate` : $\mathbb{R}^d \to \mathbb{R}$ (LayerNorm$(384) \to$ Linear$(384, 96) \to$ GELU $\to$ Linear$(96, 1)$). A frozen copy of the Stage-1 checkpoint serves as the reference policy π_{ref} (`ref_model`). Because the first 10 blocks are identical between π_θ and π_{ref}, prefix token activations are computed once under `torch.no_grad()` and branched only for the final 2 blocks, keeping peak GPU memory below 3 GB.

Let $\bar{\mathbf{a}} \in \mathbb{R}^N$ denote the head-averaged self-attention from the `[CLS]` query to all N spatial patches in the 12th layer ($\bar{a}_p = \frac{1}{H}\sum_{h=1}^{H} \mathbf{A}_{0,p}^{(h)}$), and let $\mathbf{x}_p \in \mathbb{R}^d$ be the normalized patch token embedding. We standardize $\bar{\mathbf{a}}$ per image and parameterize the independent Bernoulli foreground probability $\pi_{\theta,p} = \pi_\theta(a_p = 1 \mid I)$ for patch $p \in \{1, \ldots, N\}$ as:

$$z_p = \frac{\bar{a}_p - \mu_{\bar{a}}}{\sigma_{\bar{a}} + \epsilon}, \quad \pi_{\theta,p} = \sigma\big(s_{\text{attn}} z_p + b_{\text{spar}} + g_\phi(\mathbf{x}_p)\big), \quad \pi_{\text{ref},p} = \sigma\big(s_{\text{attn}} z_p^{\text{ref}} + b_{\text{spar}}\big) \tag{18}$$

where $\sigma(\cdot)$ is the logistic sigmoid, $s_{\text{attn}} = 1.6$ scales the standardized attention logits, $b_{\text{spar}} = \log\Big(\frac{\rho_{\text{target}}}{1-\rho_{\text{target}}}\Big)$ is a prior bias matching target foreground sparsity $\rho_{\text{target}} = 0.35$, and $g_\phi(\mathbf{x}_p)$ is the residual logit from `policy_gate` (initialized near zero so $\pi_\theta \approx \pi_{\text{ref}}$ at step 0).

2. Group Rollout Sampling and Verifiable Self-Supervised Rewards: For each training image I, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of $G = 8$ binary patch masks $\{\mathbf{a}^{(1)}, \ldots, \mathbf{a}^{(G)}\}$ where $\mathbf{a}^{(g)} \in \{0,1\}^N$ and $a_p^{(g)} \sim \text{Bernoulli}(\pi_{\theta_{\text{old}},p})$. In our strictly label-free mode (`self_supervised_v1`), each rollout $\mathbf{a}^{(g)}$ receives a deterministic verifiable reward $R(\mathbf{a}^{(g)}, I) \in [0, 1]$ composed of three synergistic geometric and spectral terms:

1. **Geometric Equivariance & Salience Consistency (R_{equiv}, weight $w_{\text{eq}} = 0.45$):** A genuine foreground object mask must transform equivariantly under horizontal reflection T. We pass the horizontally flipped image $T(I)$ through the backbone, un-flip its resulting 2D probability map $T^{-1}(\pi_\theta(T(I)))$, and blend it equally ($0.50 : 0.50$) with an unsupervised spectral-contour salience pseudo-mask $\mathbf{m}_{\text{sal}} \in \{0,1\}^N$ (derived from Sobel edge energy, border color contrast, and spectral background dissimilarity against image-border tokens). Thresholding the top $k_{\text{fg}} = \lfloor \rho_{\text{target}} N \rfloor$ patches yields binary target $\mathbf{m}_{\text{eq}}$, and we measure Intersection-over-Union:

$$R_{\text{equiv}}(\mathbf{a}^{(g)}, I) = \frac{\sum_{p=1}^{N} a_p^{(g)} m_{\text{eq},p}}{\sum_{p=1}^{N} \Big(a_p^{(g)} + m_{\text{eq},p} - a_p^{(g)} m_{\text{eq},p}\Big) + \epsilon} \tag{19}$$

2. **TokenCut Spectral & 2D Spatial Compactness (R_{compact}, weight $w_{\text{comp}} = 0.35$):** Inspired by graph-spectral TokenCut [21], we construct the non-negative patch cosine affinity matrix $\mathbf{W}_{pq} = \max(0, \hat{\mathbf{x}}_p^\top \hat{\mathbf{x}}_q)$ where $\hat{\mathbf{x}}_p = \mathbf{x}_p / \|\mathbf{x}_p\|_2$. A high-quality mask maximizes intra-foreground cohesion while minimizing the foreground–background cut and 2D spatial fragmentation (Total Variation TV_H, TV_W on the $H_g \times W_g$ grid):

$$S_{\text{NCut}}(\mathbf{a}^{(g)}) = \frac{1}{2}\left(\frac{\mathbf{a}^{(g)\top}\mathbf{W}\mathbf{a}^{(g)} - \mathbf{a}^{(g)\top}\mathbf{W}(\mathbf{1} - \mathbf{a}^{(g)})}{\mathbf{a}^{(g)\top}\mathbf{W}\mathbf{1} + \epsilon} + 1\right),$$
$$R_{\text{compact}}(\mathbf{a}^{(g)}) = 0.65\, S_{\text{NCut}}(\mathbf{a}^{(g)}) + 0.35\, \big(1 - \tfrac{1}{2}(\text{TV}_H + \text{TV}_W)\big) \tag{20}$$

3. **Target Sparsity Ratio Prior (R_{spar}, weight $w_{\text{spar}} = 0.20$):** To prevent degenerate all-foreground or empty masks, we penalize deviations of the active patch ratio $\bar{a}^{(g)} = \frac{1}{N}\sum_{p=1}^{N} a_p^{(g)}$ from $\rho_{\text{target}} = 0.35$:

$$R_{\text{spar}}(\mathbf{a}^{(g)}) = \max\left(0,\ 1 - \left(\frac{|\bar{a}^{(g)} - \rho_{\text{target}}|}{\max(\rho_{\text{target}}, 1 - \rho_{\text{target}})}\right)^{1.5}\right) \tag{21}$$

The total verifiable reward is $R(\mathbf{a}^{(g)}, I) = w_{\text{eq}} R_{\text{equiv}} + w_{\text{comp}} R_{\text{compact}} + w_{\text{spar}} R_{\text{spar}}$.

3. Unbiased Advantage, Asymmetric Clipping, and Stage-2 Composite Loss: Following Dr. GRPO [19], we compute an unbiased group-centered advantage without dividing by the group standard deviation (which otherwise inflates gradients on low-variance groups):

$$A^{(g)} = R(\mathbf{a}^{(g)}, I) - \frac{1}{G} \sum_{j=1}^{G} R(\mathbf{a}^{(j)}, I) \tag{22}$$

To eliminate sequence-length bias across N patches and couple policy optimization directly to the `[CLS]` representation $\mathbf{f}_{\text{cls}}$ (and the Value/MLP projections of the unfrozen blocks), we define the length-normalized log-probability augmented with differentiable gated-`[CLS]` cosine alignment $\mathcal{C}_{\text{align}}$:

$$\log \tilde{\pi}_\theta(\mathbf{a}^{(g)} \mid I) = \frac{1}{N} \sum_{p=1}^{N} \Big[a_p^{(g)} \log \pi_{\theta,p} + (1 - a_p^{(g)}) \log(1 - \pi_{\theta,p}) \Big] + 0.1\, \mathcal{C}_{\text{align}}(\mathbf{f}_{\text{cls}}, \mathbf{a}^{(g)}) \tag{23}$$

where $\mathcal{C}_{\text{align}}(\mathbf{f}_{\text{cls}}, \mathbf{a}^{(g)}) = \langle \hat{\mathbf{f}}_{\text{cls}}, \hat{\mathbf{f}}_{\text{fg}}^{(g)} \rangle + 0.5 \langle \hat{\mathbf{f}}_{\text{cls}}, \hat{\mathbf{f}}_{\text{cls}}^{\text{ref}} \rangle$ and $\mathbf{f}_{\text{fg}}^{(g)} = \frac{\sum_p a_p^{(g)} \mathbf{x}_p}{\sum_p a_p^{(g)}}$. With probability ratio $\rho^{(g)} = \exp\big(\log \tilde{\pi}_\theta(\mathbf{a}^{(g)} \mid I) - \log \pi_{\theta_{\text{old}}}(\mathbf{a}^{(g)} \mid I) \big)$, the Dr. GRPO loss employs DAPO-style asymmetric clipping [20] ($\epsilon_{\text{low}} = 0.20, \epsilon_{\text{high}} = 0.28$) and an exact analytical Bernoulli KL divergence penalty ($\beta_{\text{KL}} = 0.04$):

$$\mathcal{L}_{\text{Dr.GRPO}} = -\frac{1}{G} \sum_{g=1}^{G} \min\Big(\rho^{(g)} A^{(g)},\ \text{clip}\big(\rho^{(g)}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}\big) A^{(g)} \Big) + \beta_{\text{KL}}\, \mathbb{D}_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \tag{24}$$

where $\mathbb{D}_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) = \frac{1}{N} \sum_{p=1}^{N} \Big[\pi_{\theta,p} \log \frac{\pi_{\theta,p}}{\pi_{\text{ref},p}} + (1 - \pi_{\theta,p}) \log \frac{1 - \pi_{\theta,p}}{1 - \pi_{\text{ref},p}} \Big]$. Finally, we combine $\mathcal{L}_{\text{Dr.GRPO}}$ with a self-supervised foreground-guided distillation term $\mathcal{L}_{\text{FG-Distill}} = 1 - \langle \hat{\mathbf{f}}_{\text{cls}}, \hat{\mathbf{f}}_{\text{target}} \rangle$ (anchoring $\hat{\mathbf{f}}_{\text{cls}}$ to $0.88\, \hat{\mathbf{f}}_{\text{cls}}^{\text{ref}} + 0.12\, \hat{\mathbf{f}}_{\text{fg}}^{\star}$ from the highest-reward rollout $g^\star = \arg\max_g R^{(g)}$) and a binary cross-entropy gate alignment loss $\mathcal{L}_{\text{Gate-Align}}$ toward positive-advantage rollouts:

$$\mathcal{L}_{\text{Stage-2}} = \mathcal{L}_{\text{Dr.GRPO}} + 0.30\, \mathcal{L}_{\text{FG-Distill}} + 0.25\, \mathcal{L}_{\text{Gate-Align}} \tag{25}$$

During inference and evaluation, the refined representation combines the `[CLS]` embedding with policy-gated foreground pooling ($\mathbf{f}_{\text{eval}} = 0.85\, \mathbf{f}_{\text{cls}} + 0.15\, \mathbf{f}_{\text{fg}}$), and the last-layer `[CLS]` self-attention map is modulated by $\pi_{\theta,p}$ to yield sharp foreground masks.

3.5 Modular Registry and Declarative Configuration Builder

To support rigorous experimental control and reproducible ablation studies across both stages, we built a modular architecture engine consisting of:

- **Component Registry:** A decorator-based system (`@REGISTRY.register()`) decoupling class implementations from network construction. Global registries include `MODULES`, `BACKBONES`, `HEADS`, `MODELS`, `LOSSES`, `OPTIMIZERS`, and `TRANSFORMS` (registering both Stage-1 modules and Stage-2 `DINOGRPOPolicyWrapper` and `DrGRPOLoss`).
- **Declarative Configuration Builder:** A unified builder pipeline (`build_model`, `build_loss`, `build_optimizer`, and `build_schedulers`) that instantiates paired Student–Teacher networks, multi-crop wrappers, and cosine schedules directly from human-readable YAML configurations without altering training scripts.

4 Experimental Setup

4.1 Pretraining, Post-Training, and Evaluation Pipeline

Following the standard evaluation benchmark of DINOv1 [1], our experimental workflow consists of two strictly label-free training stages on ImageNet-100 followed by frozen-backbone evaluation on CIFAR-10:

- **Stage-1 Unlabeled Self-Supervised Pretraining on ImageNet-100:** Self-supervised pretraining (`train.py`) is conducted strictly without class labels on **ImageNet-100** (`clane9/imagenet-100`, a 100-class subset of ImageNet-1K [16, 17] containing ∼130,000 training images) for **190 epochs** on an **NVIDIA H100 GPU**. During pretraining, each unlabeled image is transformed via the DINO multi-crop pipeline (`DataAugmentationDINO`):
  - **Global Crops:** 2 views sampled with area scale $[0.4, 1.0]$ and resized via bicubic interpolation to 224×224 pixels (14×14 patch grid). Both Teacher and Student networks process these global views.
  - **Local Crops:** 6 views sampled with area scale $[0.05, 0.4]$ and resized to 96×96 pixels (6×6 patch grid). Only the Student network processes local views.
  - **Photometric Augmentations:** Sequential application of random horizontal flipping ($p = 0.5$), color jittering (brightness 0.4, contrast 0.4, saturation 0.2, hue 0.1 with $p = 0.8$), random grayscale ($p = 0.2$), Gaussian blur ($p = 1.0$ for global crop 1, $p = 0.1$ for global crop 2, $p = 0.5$ for local crops), and solarization ($p = 0.2$ on global crop 2).
- **Stage-2 Self-Supervised Dr. GRPO Post-Training on ImageNet-100:** Starting from the Stage-1 epoch-190 checkpoints of both Vanilla `vit_small` and proposed `vit_small_sparse_rope`, we execute 10 epochs of label-free RLVR post-training (`train_grpo_post.py`) on a class-balanced stratified subset of 5,000 unlabeled ImageNet-100 training images (batch size 64, 79 steps/epoch) on an **NVIDIA Tesla T4 GPU** (16 GB VRAM). We use dual-rate AdamW optimization: a conservative learning rate ($0.25 \times \text{lr} = 2 \times 10^{-5}$) for the last 2 ViT blocks to preserve pretrained representations, and a fast learning rate ($5.0 \times \text{lr} = 4 \times 10^{-4}$) for the newly initialized `policy_gate`.
- **Downstream Evaluation on CIFAR-10:** After training on ImageNet-100, we freeze the models and evaluate the learned visual representations on the full **CIFAR-10** benchmark [14] (`eval/eval_knn.py`, `eval/eval_linear.py`, and `eval/eval_ablation_2x2.py`), comprising 50,000 training images and 10,000 test images across 10 semantic categories (*airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck*). During evaluation, images are resized to 256×256, center-cropped to 224×224, and normalized using standard ImageNet statistics.

4.2 Hardware and Hyperparameter Specifications

Stage-1 pretraining was executed on an **NVIDIA H100 GPU** for **190 epochs**, while Stage-2 Dr. GRPO post-training was executed for **10 epochs** using PyTorch Automatic Mixed Precision (AMP). Full architectural and training hyperparameters for both stages are summarized in Table 1.

4.3 Evaluation Protocols

Following standard DINO evaluation conventions [1] and unsupervised object discovery benchmarks [21], we evaluate frozen models on CIFAR-10 across four complementary metrics:

1. **k-Nearest Neighbors (k-NN) Classifier:** The backbone is frozen and L2-normalized features are extracted for all 50,000 training and 10,000 test samples of CIFAR-10. Top-1 and Top-5 accuracy are computed via weighted cosine voting ($T = 0.07$ at $k \in \{10, 20, 200\}$).

Table 1: Stage-1 Pretraining and Stage-2 Dr. GRPO Post-Training Hyperparameters.

Hyperparameter	**Value / Specification**
Stage-1: DINOv1 Self-Supervised Pretraining (190 Epochs)	
Backbone Architecture	Vision Transformer Small (`ViT-Small/16`, 12 layers, $d = 384$, $H = 6$)
Positional Embedding	Frequency-Adaptive 2D RoPE (`adaptive_2d`, base 10000.0)
Sparse Attention Settings	$K_{\text{local}} = 3$, $K_{\text{semantic}} = 16$, Bidirectional `[CLS]` Routing
Projection Head (DINOHead)	3-layer MLP (hidden 2048, bottleneck 256, out 16384)
Pretraining Dataset & Hardware	Unlabeled **ImageNet-100** ($\sim$130k images) on **NVIDIA H100 GPU**
Effective Batch Size	128 (Local BS = 32, Gradient Accumulation = 4, AMP enabled)
Stage-1 Optimizer & LR	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$), Cosine LR $5 \times 10^{-4} \rightarrow 1 \times 10^{-5}$ (10 warmup epochs)
Temperatures & Momentum	$\tau_t = 0.04$, $\tau_s = 0.10$, EMA momentum $\lambda : 0.996 \rightarrow 1.000$
Stage-2: Self-Supervised Dr. GRPO Post-Training (10 Epochs)	
Policy Wrapper (`DINOGRPOPolicyWrapper`)	Frozen Blocks $0 \ldots 9$ + Unfrozen Blocks $10, 11$ + `policy_gate` ($384 \rightarrow 96 \rightarrow 1$)
Stage-2 Trainable Parameters	3,587,521 / 43,369,153 total wrapper params (**8.27%**, < 3 GB VRAM)
Group Rollout Size (G)	$G = 8$ binary patch masks sampled per image ($N = 196$ patches)
Verifiable Reward Mode	`self_supervised_v1`: $w_{\text{eq}} = 0.45$, $w_{\text{comp}} = 0.35$, $w_{\text{spar}} = 0.20$
Target Foreground Sparsity	$\rho_{\text{target}} = 0.35$ (Initial logit bias $b_{\text{spar}} = -0.619$, scale $s_{\text{attn}} = 1.6$)
Dr. GRPO Clipping & KL Anchor	Asymmetric $\epsilon_{\text{low}} = 0.20, \epsilon_{\text{high}} = 0.28$, Analytical $\beta_{\text{KL}} = 0.04$
Stage-2 Optimizer & Dual LRs	AdamW (Base lr $= 8 \times 10^{-5}$: Backbone 2×10^{-5}, Gate 4×10^{-4}, Cosine schedule)

2. **Linear Probing:** A linear classifier ($384 \rightarrow 10$) is trained on frozen features using SGD with momentum 0.9, cosine annealing learning rate, and batch size 256 (100 epochs on standardized features for the Stage-1 main benchmark; 30 epochs with lr $= 0.1$ on raw features in the unified 2×2 ablation suite).

3. **Unsupervised Foreground Mask IoU (%):** Thresholding the top $\lfloor \rho_{\text{target}} N \rfloor$ patches of the last-layer `[CLS]` attention map yields a binary foreground prediction $\hat{\mathbf{m}} \in \{0, 1\}^N$, which is evaluated via mean Intersection-over-Union against the unbiased spectral-contour foreground mask across all 10,000 CIFAR-10 test images.

4. **Correct Localization (`CorLoc`, %):** The percentage of test images in which the predicted unsupervised foreground mask achieves IoU ≥ 0.50 with the target object mask.

5 Results and Empirical Analysis

5.1 Quantitative Classification Performance

Table 2 compares our proposed architecture against the standard ViT-Small baseline on CIFAR-10 after 190 epochs of self-supervised pretraining on ImageNet-100.

Key Observations:

Table 2: Classification benchmark on CIFAR-10 using checkpoints pretrained for 190 epochs on ImageNet-100.

Evaluation Metric	Default ViT-Small	RoPE + Sparse	Gain (Δ)
k-NN Top-1 ($k = 10$)	74.41%	**75.97%**	**+1.56%**
k-NN Top-5 ($k = 10$)	96.23%	**96.55%**	**+0.32%**
k-NN Top-1 ($k = 20$)	74.34%	**75.31%**	**+0.97%**
k-NN Top-5 ($k = 20$)	97.42%	**97.79%**	**+0.37%**
k-NN Top-1 ($k = 200$)	70.41%	**70.65%**	**+0.24%**
Linear Probe Top-1 (Epoch 10)	80.92%	**83.38%**	**+2.46%**
Linear Probe Top-1 (Epoch 20)	81.09%	**83.84%**	**+2.75%**
Linear Probe Top-1 (Epoch 30)	80.85%	**83.63%**	**+2.78%**
Linear Probe Top-1 (Epoch 50)	80.98%	**83.70%**	**+2.72%**
Linear Probe Top-1 (Epoch 100)	81.34%	**83.90%**	**+2.56%**
Linear Probe Top-1 (**Best Checkpoint**)	81.46%	**83.98%**	**+2.52%**
Linear Probe Top-5 (**Best Checkpoint**)	99.31%	**99.44%**	**+0.13%**
Linear Probe Final Train Loss (Epoch 100)	0.4566	**0.4019**	**-0.0547**
Linear Probe Final Val Loss (Epoch 100)	0.5457	**0.4770**	**-0.0687**

- **Consistent Representation Superiority:** Our model outperforms the baseline across all metrics, achieving **75.97%** ($k = 10$) and **75.31%** ($k = 20$) k-NN Top-1 accuracy (+1.56% and +0.97%), and **83.98%** Best Linear Probe Top-1 accuracy (+2.52% over 81.46%).
- **Rapid Linear Convergence:** By epoch 10 of linear probing, our model already achieves **83.38%** Top-1 accuracy (+2.46% over the baseline's 80.92% at epoch 10, and already exceeding the baseline's 100-epoch best of 81.46%).
- **Lower Optimization and Validation Loss:** Both the final training loss (0.4019 vs. 0.4566) and validation loss (0.4770 vs. 0.5457) are substantially lower, confirming tighter within-class clustering and wider margin separation.

5.2 Parameter Complexity and Hardware Profiling

To verify whether performance improvements stem from architectural inductive bias rather than model capacity inflation, we profiled parameters, floating-point operations (FLOPs), latency, and throughput on an RTX 3070 GPU.

Table 3: Model Parameters, Theoretical Complexity, and Hardware Profiling (224×224 resolution).

Metric / Resource	Default ViT-Small	RoPE + Sparse	Comparison
Backbone Trainable Params	21,665,664	21,665,664	**0% Overhead (Identical)**
Non-trainable Buffers	0	32 elements	Frequency cache (`inv_freq`)
Projection Head Params	9,720,064	9,720,064	Identical
Total Model Parameters	**31,385,728**	**31,385,728**	**31.39M (Identical)**
Backbone FLOPs (224×224)	9.20 GFLOPs	9.20 GFLOPs	**Exact Parity (4.60 GMACs)**
DINO Head FLOPs	0.019 GFLOPs	0.019 GFLOPs	Identical
Total FLOPs per Image	**9.22 GFLOPs**	**9.22 GFLOPs**	**Exact Parity**
Inference Latency (BS=1)	**1.20 ms**	2.20 ms	+1.00 ms (RoPE rotation)
Throughput (BS=128, img/s)	**831.9**	455.0	Memory bandwidth trade-off

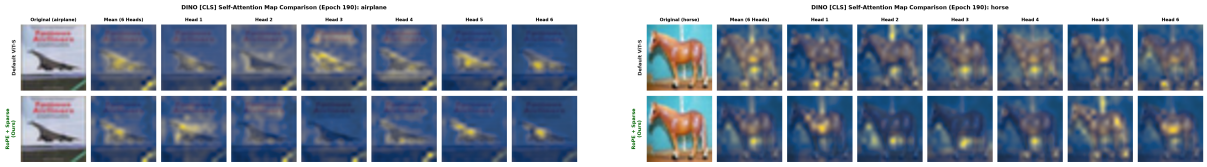
As shown in Table 3, the proposed improvements introduce **zero additional trainable parameters** (exactly 21.67M backbone parameters). The theoretical FLOP count is identical at 9.20 GFLOPs,

Table 4: 2×2 Ablation Study across Architecture (`vit_small` vs. `vit_small_sparse_rope`) and Stage-2 RLVR Post-Training (`Dr. GRPO`) on the full CIFAR-10 benchmark (50,000 train / 10,000 test images).

Model Variant	Backbone Arch	Stage-2 RLVR	k-NN Top-1	k-NN Top-5	Linear Probe	Foreground IoU	CorLoc (≥ 0.5)
1. Vanilla DINOv1	`vit_small`	No	71.13%	96.57%	78.07%	58.88%	79.93%
2. RoPE+Sparse DINOv1	`vit_small_sparse_rope`	No	71.83%	96.98%	80.22%	55.45%	69.26%
3. Vanilla + Dr. GRPO	`vit_small`	Dr. GRPO	71.29%	96.73%	78.44%	**78.98%**	**99.91%**
4. RoPE+Sparse + Dr. GRPO (Ours)	`vit_small_sparse_rope`	**Dr. GRPO**	**72.05%**	**97.12%**	**80.67%**	**72.44%**	**98.17%**
Effect of Dr. GRPO on Vanilla DINOv1 (Variant 3 vs. 1)			**+0.16%**	**+0.16%**	**+0.37%**	**+20.10%**	**+19.98%**
Effect of Dr. GRPO on RoPE+Sparse (Variant 4 vs. 2)			**+0.22%**	**+0.14%**	**+0.45%**	**+16.99%**	**+28.91%**
Full Proposed Method vs. Vanilla Baseline (Variant 4 vs. 1)			**+0.92%**	**+0.55%**	**+2.60%**	**+13.56%**	**+18.24%**

confirming that accuracy gains arise purely from structured geometric representation rather than increased capacity. The minor increase in latency (1.20 ms $\rightarrow$ 2.20 ms) is attributed to element-wise rotary trigonometric multiplications and Top-K masking operations in standard PyTorch, which can be further optimized via custom Triton kernels.

5.3 Qualitative Analysis: Attention Map Visualizations



(a) Airplane class (b) Horse class

Figure 4: Visualization of Self-Attention maps for the `[CLS]` token across all 6 heads in the final Transformer layer on CIFAR-10 samples. Baseline ViT-Small exhibits background noise leakage, whereas our RoPE + Sparse Attention model exhibits sharp foreground localization.

We inspect the self-attention maps of the `[CLS]` token with respect to spatial patch tokens in the 12th Transformer block across all 6 attention heads:

1. Airplane Class (Figure 4a): In the baseline ViT model, attention is diffusely scattered across the sky background (notably in Heads 1 and 3). In contrast, our RoPE + Sparse Attention model focuses tightly on the aircraft fuselage, wings, and turbine engines. Heads 2 and 5 delineate distinct mechanical sub-structures while completely ignoring sky distractors.

2. Horse Class (Figure 4b): In the baseline, attention responds to general color tones, causing the horse's silhouette to bleed into the grassy background. In our model, attention maps trace the horse's curved back, head, and legs. The high-frequency components of the Frequency-Adaptive 2D RoPE preserve fine-grained spatial coordinates, enabling the network to resolve clear object-background boundaries.

5.4 2×2 Ablation Study: Architecture $\times$ Stage-2 Dr. GRPO Post-Training

To isolate the individual and combined contributions of our Stage-1 architectural design (`vit_small_sparse_rope`) and Stage-2 self-supervised RLVR post-training (`Dr. GRPO`), we conduct a controlled 2×2 ablation study across all four model variants on the full CIFAR-10 dataset (50,000 training / 10,000 test images) using `eval/eval_ablation_2x2.py`. Table 4 reports downstream classification (k-NN at $k = 20$ and 30-epoch unnormalized Linear Probe with lr $= 0.1$) alongside unsupervised object localization (Foreground Mask IoU and `CorLoc` at IoU ≥ 0.50).

Table 5: Stage-2 Dr. GRPO Training Dynamics over 10 Self-Supervised RLVR Epochs on ImageNet-100 (evaluated at each epoch on CIFAR-10 validation split).

Epoch	Vanilla + Dr. GRPO (`def_grpo`)				RoPE+Sparse + Dr. GRPO (`rop_grpo`)			
	Reward	**KL Div**	**k-NN (%)**	**FG IoU (%)**	**Reward**	**KL Div**	**k-NN (%)**	**FG IoU (%)**
0 (Pre-RL)	—	0.0000	62.85	58.72	—	0.0000	64.90	55.59
1	0.6613	0.1066	63.05	76.32	0.6045	0.1172	65.20	71.17
2	0.7330	0.1980	63.25	77.42	0.6561	0.2116	**65.40**	71.82
4	0.7348	0.2040	63.40	77.74	0.6711	0.2420	65.10	72.31
6	0.7403	0.2104	**63.45**	78.33	0.6782	0.2557	65.15	71.88
8	0.7423	0.2142	63.30	78.71	0.6804	0.2621	65.25	71.78
10	**0.7443**	0.2178	63.40	**78.82**	**0.6826**	0.2650	65.20	**71.56**

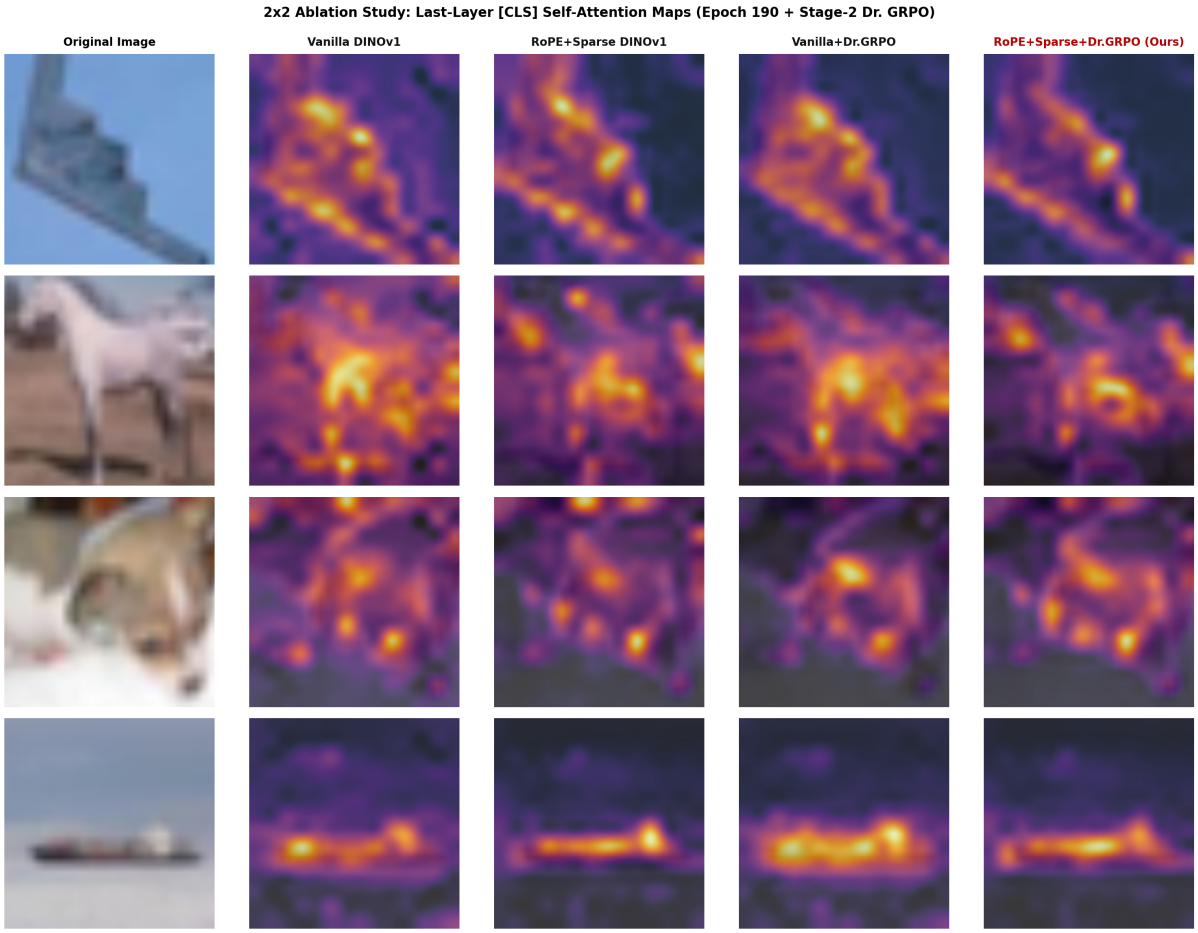


Figure 5: Qualitative 2×2 Ablation Comparison of last-layer `[CLS]` self-attention heatmaps across four representative CIFAR-10 categories (*airplane, horse, dog, ship*). From left to right: (1) Original Image, (2) Stage-1 Vanilla DINOv1, (3) Stage-1 RoPE+Sparse DINOv1, (4) Stage-2 Vanilla + Dr. GRPO, and (5) Stage-2 RoPE+Sparse + Dr. GRPO (Ours). Stage-2 RLVR post-training suppresses residual background noise and concentrates attention mass tightly within true object contours.

Analysis of 2×2 Ablation Findings:

- **Massive Improvement in Unsupervised Object Localization:** Before RLVR post-training, Stage-1 RoPE+Sparse concentrates attention on discriminative object sub-parts (e.g., animal heads, aircraft wings), which improves classification (+2.15% Linear Probe over Vanilla) but yields a slightly lower holistic mask IoU (55.45% vs. 58.88%). Applying Stage-2 Dr. GRPO resolves this trade-off: Fore-

ground Mask IoU surges by **+20.10%** on Vanilla DINOv1 ($58.88\% \rightarrow 78.98\%$) and by **+16.99%** on RoPE+Sparse DINOv1 ($55.45\% \rightarrow 72.44\%$), while `CorLoc` jumps from 69.26% to **98.17%** (+28.91%) on our RoPE+Sparse backbone and reaches **99.91%** on Vanilla+Dr. GRPO.

- **Synergy Between Geometric Pretraining and RLVR Post-Training:** Unlike naive fine-tuning that often degrades pretrained features (catastrophic forgetting), our analytical Bernoulli KL anchor ($\beta_{\text{KL}} = 0.04$) and foreground-guided teacher distillation ($\mathcal{L}_{\text{FG-Distill}}$) simultaneously improve downstream classification. Our full model (`RoPE+Sparse+Dr.GRPO`) achieves the highest classification performance across all four variants: **72.05%** k-NN Top-1 (+0.92% over Vanilla), **97.12%** k-NN Top-5, and **80.67%** 30-epoch Linear Probe accuracy (+2.60% over Vanilla).
- **Fast, Monotonic RLVR Convergence (Table 5):** Across the 10 post-training epochs on ImageNet-100, the mean group reward rises monotonically from $0.6613 \rightarrow 0.7443$ (`def_grpo`) and $0.6045 \rightarrow 0.6826$ (`rop_grpo`), while the KL divergence stabilizes cleanly around 0.21–0.26, confirming stable policy improvement without policy collapse.
- **Visual Verification (Figure 5):** The 5-column visual grid confirms that while Stage-1 models exhibit peripheral attention halos in complex scenes, Stage-2 Dr. GRPO variants cleanly segment the full body of the airplane, horse, dog, and ship with near-zero background activation.

6 Ablation Studies and Discussion

6.1 Coordinate Adaptation Modes in 2D RoPE

We examined four modes for frequency coordinate adaptation:

- `standard_2d` ($\alpha_m = 1.0$): Raw coordinates across all channels. While effective for fixed resolutions, transfer across multi-scale crops exhibits slight degradation due to high-frequency domain mismatch.
- `normalized_2d` ($\alpha_m = 0.0$): Fully normalized coordinates. This mode accommodates resolution shifts but diminishes fine spatial texture sensitivity.
- `adaptive_2d` (Proposed): Linear frequency ramp. This mode achieves the highest overall accuracy (**75.31%** k-NN at $k = 20$, **75.97%** at $k = 10$), balancing local detail preservation and global scale invariance.
- `learnable_2d`: Parameterized α_m via unconstrained logit sigmoid. Yields comparable accuracy to `adaptive_2d` while requiring additional optimization tuning.

6.2 Sparsity Factor vs. Representation Expressiveness

Restricting attention to a 3×3 local window and $K_{\text{semantic}} = 16$ tokens reduces the active key set per spatial query to $9+16+1 = 26$ tokens out of 197 total tokens (an 86.8% reduction in active attention pairs). The fact that this configuration outperforms dense attention (+2.52% in Linear Probing) demonstrates that dense visual attention contains substantial redundancy, and that inductive spatial-semantic filtering acts as an effective regularizer against background overfitting.

6.3 Design Choices in Stage-2 Dr. GRPO

Our Stage-2 post-training design incorporates three critical mechanisms that prevent trivial mask degeneration: (1) **Unbiased Dr. GRPO Advantage:** Omitting group standard-deviation normalization prevents gradient explosions when all $G = 8$ sampled masks on a high-confidence image yield similar rewards; (2) **Shared-Prefix Partial Unfreezing:** Freezing the first 10 ViT blocks preserves low- and mid-level visual primitives learned over 190 epochs on ImageNet-100 while reducing trainable parameters to 8.27%; and

(3) **Verifiable Tri-Objective Reward:** Combining horizontal-flip equivariance ($w_{\text{eq}} = 0.45$), spectral TokenCut compactness ($w_{\text{comp}} = 0.35$), and target sparsity ($w_{\text{spar}} = 0.20$) ensures that the policy cannot game the reward by predicting either a single salient patch or the entire image grid.

7 Conclusion

In this paper, we presented a two-stage self-supervised Vision Transformer framework for DINOv1 that unifies **Frequency-Adaptive 2D Rotary Position Embeddings (2D RoPE)** and **Spatial-Semantic Sparse Attention** in Stage-1 pretraining with **Unbiased Group Relative Policy Optimization (Dr. GRPO)** in Stage-2 post-training. In Stage 1, decoupling spatial frequencies and filtering dense attention into local, semantic, and bidirectional `[CLS]` pathways yields consistent improvements on CIFAR-10 (**75.97%** k-NN at $k = 10$, **75.31%** at $k = 20$, and **83.98%** standardized Linear Probing Top-1 accuracy, +2.52% over the baseline) with zero parameter overhead (21.67M backbone parameters) and exact theoretical FLOP parity (9.20 GFLOPs). In Stage 2, formulating foreground patch gating as a label-free RLVR problem with verifiable geometric equivariance and TokenCut spectral rewards dramatically enhances unsupervised object localization—boosting Foreground Mask IoU by **+16.99% to +20.10%** and achieving **98.17%–99.91%** `CorLoc` across our 2×2 CIFAR-10 ablation study while simultaneously improving downstream classification.

Future work will investigate scaling this two-stage framework to full ImageNet-1K / ImageNet-22K pretraining, integrating register tokens [9], implementing specialized Triton sparse kernels to recover wall-clock throughput parity, and extending Stage-2 verifiable visual rewards to dense multi-object segmentation and video tracking benchmarks.

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